

PANINI UI/UX Design System: The Algorithmic Interface

1. Vision & Philosophy

Moving beyond the standard "Sci-Fi Blue JARVIS," PANINI embraces "Algorithmic Antiquity." Inspired by the world's first formal algorithmic structures crafted by Pāṇini, this design system merges ancient geometric precision with Google Material Design 3 principles (adaptive elevation, state layers, and choreographed motion).

The interface is not just a tool; it is a living derivation tree. It utilizes Three.js for 3D state representations, providing a tactile, grounded experience for a powerful multi-agent system.

2. Color Palette: The Digital Sutra

Instead of neon, PANINI uses a deeply grounded, high-contrast dark mode palette.

- Obsidian Base (#0B0C10): The deepest background, allowing glowing elements to pop without eye strain.
- Charcoal Surface (#1F2833): Used for elevated cards and containment (Material Elevation).

- Copper Primary (#B87333): The core brand color. Represents the "Sutra" (thread). Used for primary actions, active nodes, and the Electron avatar.
- Amber Accent (#FFBF00): Used for active processing, warnings, and highlighting derivation paths.
- Gold/Sand Text (#E5C158): High-legibility text for primary readouts.

3. Typography

- Primary/Headers: Google Sans or Outfit. Clean, geometric, and modern.
- `Runtime & Telemetry (Code): JetBrains Mono. Used for the gateway, backend logs, and the Code capability workspace.`
- Accents: Noto Sans Devanagari. Used subtly for watermarks or background module titles (e.g., "अन्वेषण" faintly behind the Search module).

4. Logo & Iconography

The Algorithmic P:
The PANINI logo represents a linguistic node network shaping the letter 'P'.
Iconography System:
Icons follow Material Design's line-art style but are rendered with a 1px copper glow.

- Search: A radiating node.
- Browser: A geometric globe/web.
- Desktop: A wireframe window cascade.
- Code: A derivation tree (AST).

5. The Electron Avatar (Desktop Controller)

The avatar is a constant, unintrusive presence on the Windows desktop. Built with a transparent Electron window and Three.js.

- State 0 | Idle: A minimalist, glowing copper core (Sri Yantra or Icosahedron) pulsing at 12 BPM (resting human breath).
- State 1 | Listening: The core flattens into concentric rings. Audio waveforms ripple outward like water drops, abandoning standard bar equalizers.
- State 2 | Delegating: When a command is routed to a Capability, the core fragments. A glowing "Sutra" thread shoots toward the edge of the screen corresponding to the capability (e.g., bottom-left for Browser).

6. The Web Console (localhost:3200) Screen Designs

The Web Console is the operational dashboard. It applies Google's Material Grid system.

Screen: The Dashboard Overview

- Layout: A 3-column masonry layout.
- Center Stage (The Engine): A real-time 3D node-graph visualizer. When a user speaks, the root intent is shown, and branches sprout as the system delegates to Search, Browser, Desktop, and Code.
- `Left Panel (Telemetry): A unified stream of events from the Runtime Gateway. Uses a sticky header with a subtle backdrop-filter (blur).`
- Right Panel (Capabilities Grid): Four interactive cards for Search, Browser, Desktop, and Code.

Screen: Active Operation - "Code"

- Interaction: When the Code agent is invoked, the Code card expands using a Shared Axis transition (Material Motion), taking over the Right Panel.
- `Visuals: The background of the card shifts to deep Charcoal. A glowing copper border indicates the agent is writing/running code in the approved workspace. A terminal window displays real-time execution in JetBrains Mono.`

Screen: Active Operation - "Browser" (JevRev Claw)

- Interaction: The Browser card elevates (Material Shadow level 4).
- Visuals: Displays a minimized picture-in-picture view of the Chrome tab being manipulated, overlaid with a wireframe highlighting the DOM elements the Claw is currently interacting with.

Screen: Active Operation - "Desktop"

- Interaction: The Desktop card utilizes state layers to show active observation.
- `Visuals: Displays a named pipe event stream showing exact Windows API calls in a beautiful, cascading waterfall UI.`

7. Motion & Interaction Principles

- Choreography: UI elements should not just appear; they should "derive." Lists populate sequentially to mimic algorithmic generation.
- Haptic Feedback (Visual): Since it's a visual interface, successful actions (like code tests passing) result in a subtle amber flash across the component's border, rewarding the user.

AVATAR DESIGN IDEATION DOCUMENT: PANINI

1. INTRODUCTION

This document collates all architectural and behavioral design explorations for the PANINI Electron avatar. PANINI is a voice-first Windows assistant, rooted in the philosophical concept of **"Algorithmic Antiquity."**
Instead of traditional sci-fi glowing interfaces, the PANINI aesthetic bridges ancient structural logic (the Aṣṭādhyāyī rules) with modern holographic and particle visualization. The avatar acts as the central **Controller and Multi-Agent Gateway**, visually delegating tasks to specific capabilities: Search, Browser (JevRev Claw), Desktop (Supported Apps), and Code.

2. THE VISUAL LANGUAGE: ALGORITHMIC ANTIQUITY

This philosophy dictates the foundational design tokens across all avatar concepts.

- **Core Palette:** Copper (Primary), Amber/Gold (Accent), Obsidian/Charcoal (Base Surface). This palette evokes fire, ancient metallic inscriptions, and premium technological depth, standing out from standard "JARVIS blue" sci-fi.
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- **Typography:** `highly precise monospace (e.g., JetBrains Mono) for runtime logs,` geometric sans-serif for interface.
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- **Background Motifs:** faint, rotating mandalas and vertical scrolling Sanskrit characters (like matrix code, but slower and rooted in structure).
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3. CONCEPT 1: THE GEOMETRIC BASE (FUNCTIONAL ANCHOR) (SHORTLISTED FOR FINAL DESIGN)

This concept serves as the foundational geometric visualization of PANINI's logic.



- **Visualization:** A symmetrical geometric brain/node network (derived from the original 'P' logo and Sri Yantra), rendered in polished copper and translucent crystal. It represents the fundamental ruleset, or 'Dhatu'.
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- **States & Behavior:** Detailed in the original functionality specification:
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    - **Idle:** minimalist, glowing copper geometric core (Tessellation) pulsing at **12 BPM** (resting human breath).
    -
    - **Listening:** The core expands, transforming into **concentric, rippling mandala-like rings**. Standard audio bar equalizers are abandoned; waveforms ripple outward like water drops.
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    - **Delegating (Function):** The core fragmentates. Glowing "Sutra" threads are routed diagonally from the core nodes toward the edge of the screen, connecting to distinct capability nodes (Search, Code, etc.) at the Windows desktop boundary. This visualization grounds the multi-agent delegation logic.
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- **Best For:** Stability, functional grounding, and establishing the baseline design.
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4. CONCEPT 2: THE QUANTUM CYMATIC SWARM (DYNAMIC FLUID)

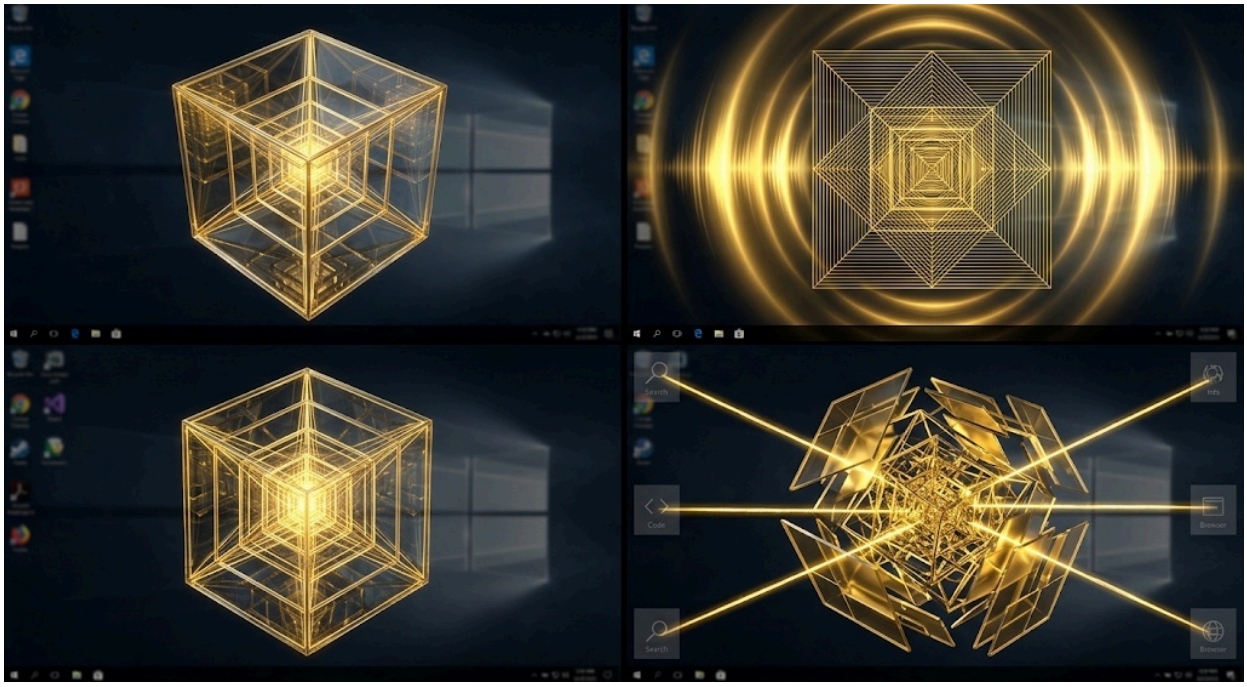
This concept pushes the boundaries of fluid behavior and auditory feedback.



- **Visualization:** Abandoning rigid wireframes, this avatar is a dense, borderless sphere of thousands of anti-gravity copper particles representing individual data points.
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- **States & Behavior:** detailed in the original dynamic specification:
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    - **Idle:** The swarm orbits a heavy, dark-matter point, resembling a trapped, dynamic hourglass.
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    - **Listening (Cymatic Mode):** Particles react directly to voice frequency, forming intricate, symmetrical **cymatic patterns** (like mandalas on a vibrating plate). Low frequency creates heavy waves; high frequency creates intricate geometry.
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    - **Delegating:** The swarm shatters. Particles rapidly flow across the desktop like a liquid river, reassembling to form the exact icon of the active capability at the screen edge.
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- **Best For:** Fluidity, complex audio reactivity, and visualizing massive, borderless data processing.
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5. CONCEPT 3: THE RECURSIVE TESSERACT (GEOMETRIC RECURSION)

This concept visualizes deep mathematical logic and the recursive nature of the Aṣṭādhyāyī rules.



- **Visualization:** A complex, glass-and-gold 4D hypercube (Tesseract) constantly folding inside out. Nested infinite cubes are visible rotating on impossible axes within the core.
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- **States & Behavior:** detailed in the original recursive specification:
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    - **Idle:** The core constantly folds and rotates recursively.
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    - **Listening:** The Tesseract abruptly stops folding and **flattens into a precise 2D geometric grid**, pulsing violently with voice like a speaker diaphragm.
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    - **Delegating:** The outer cube unlocks and physically dismantles. The exposed inner nodes shoot out powerful **laser-like 'Sutra' threads**, piercing the edges of the Windows screen to physically 'anchor' the capability modules. The central core spins chaotically.
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- **Best For:** Infinite recursive depth, mathematical visualization, and high-intensity, physical interaction (laser anchors).
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6. CONCEPT 4: THE KINETIC SUTRA RIBBON (LINGUISTIC HOLOGRAM)

This concept moves entirely away from geometric cores to embrace pure linguistic structure as form.



- **Visualization:** A live, continuous ribbon of glowing Sanskrit characters (Aṣṭādhyāyī rules) floating in 3D space, abandoning an central core.
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- **States & Behavior:** detailed in the original typographic specification:
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    - **Idle:** Tangled into a tight, smoldering **'Hot Iron Knot'** in the screen corner.
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    - **Listening:** The knot uncoils instantly. The characters form a **perfect, spinning ring** that locks onto the Windows mouse cursor, expanding and contracting with audio frequency.
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    - **Delegating:** The ribbon violently whips across the desktop in a light-painting trail. If it delegates to "Search," it wraps itself into a geometric node; for "Code," it zig-zags into the shape of glowing curly brackets { } and physically **types out the first line of the runtime log** before fading.
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- **Best For:** Unique linguistic representation, typographic holograms, and visualizing the literal application of rules (Aṣṭādhyāyī) through typing and trails.
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7. IDEATION SUMMARY TABLE

Concept	Theme	Idle Focus	Listening Focus	Delegation Focus	Key Challenge (WebGL)
Concept 1: Geometric Base	Functional	stable core pulse	concentric audio ripples	routing to desktop edges	balancing particle count and texture
Concept 2: Cymatic Swarm	Fluid	anti-gravity particle orbit	cymatic mandala formation	liquid particle reassembly	massive particle swarm physics simulation
Concept 3: Recursive Tesseract	Mathematical	4D folding hypercube	2D speaker flattening	laser thread anchoring	visualizing infinite recursion without overload
Concept 4: Sutra Ribbon	Linguistic	hot iron smoldering knot	spinning ring cursor lock	light trail & typed brackets	real-time typographic deformation and typing